

exif-turbo User Manual

exif-turbo lets you scan image folders, build a searchable index of all EXIF metadata, and instantly find any photo by camera model, lens, date, location, or any other tag — across thousands of images. Video files (MP4, MOV, AVI, MKV, WMV, M4V, MTS, M2TS, 3GP, WebM, FLV) are indexed alongside still images, with thumbnails and previews extracted from the embedded thumbnail or a frame at 1/3 of the duration.

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1. Requirements

ExifTool

exif-turbo requires **ExifTool** on your PATH to extract metadata from images.

Platform	Install
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Windows (MSI)	ExifTool is bundled with the MSI installer — no separate download needed. A system-wide <code>exiftool.exe</code> on your <code>PATH</code> takes priority if present.
Windows (source)	Download the standalone <code>.exe</code> from exiftool.org , rename to <code>exiftool.exe</code> , place in a folder on your <code>PATH</code>
macOS	<code>brew install exiftool</code> (if Homebrew is not installed yet: <code>/bin/bash -c "\$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install"</code>
Linux	<code>sudo apt install libimage-exiftool-perl</code>

2. Installation

Windows installer (recommended)

Download `exif-turbo-<version>-windows.msi` from the [Releases page](#). The installer adds an entry to **Start Menu** and puts `exif-turbo` on your `PATH`. ExifTool is **included in the MSI** — you do not need to download it separately. If you already have ExifTool installed system-wide, that version takes priority.

macOS installer

Download `exif-turbo-<version>-macos-arm64.dmg` for Apple silicon or `exif-turbo-<version>-macos-intel.dmg` for an Intel Mac from the same Releases page, open it, and drag **exif-turbo.app** into your **Applications** folder.

Linux package

Download `exif-turbo-<version>-linux-amd64.deb` or `exif-turbo-<version>-linux-arm64.deb` (Debian/Ubuntu), or `exif-turbo-<version>-linux-x86_64.rpm` (Fedora/openSUSE), from the Releases page and install it with your package manager:

```
# Debian / Ubuntu
sudo apt install ./exif-turbo-<version>-linux-amd64.deb
```

```
# Fedora / openSUSE
sudo dnf install ./exif-turbo-<version>-linux-x86_64.rpm
```

The package installs the application to `/opt/exif-turbo/`, registers a desktop launcher, and creates a `/usr/bin/exif-turbo` symlink.

From source

```
pip install -e .
```

Command-line options

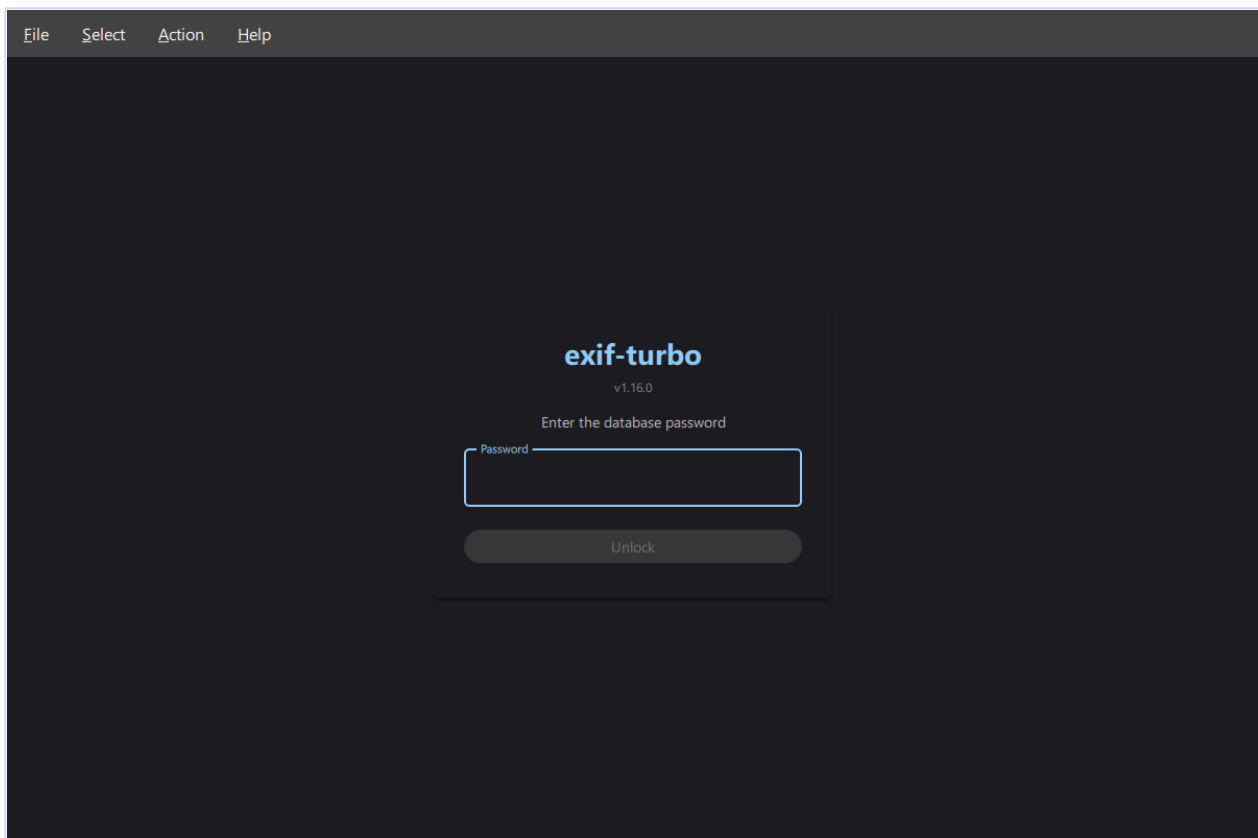
```
exif-turbo [--db NAME] [--version]
```

Option	Description
<code>--db NAME</code>	Open (or create) a named database instead of the default one. The database is always stored in <code>~/.exif-turbo/data/<NAME>/<NAME>.db</code> . Useful for keeping separate libraries — e.g. <code>exif-turbo --db work</code> and <code>exif-turbo --db holidays</code> .
<code>--version</code>	Print the installed exif-turbo version and exit.

If `--db` is omitted, the default database `~/.exif-turbo/data/index/index.db` is used.

3. First Launch — Unlocking the Database

When you start `exif-turbo` you are greeted by the **lock screen**:



First-time setup — creating a new database

If this is the first time you have launched exif-turbo (or you are using a new named database), the lock screen shows a **New passphrase** field, a **Confirm passphrase** field, and a **Create Database** button. Choose a strong passphrase of at least 12 characters that mixes letters, numbers, and symbols — it encrypts your entire image index and **cannot be recovered if lost**.

Opening an existing database

Enter your password in the **Password** field and click **Unlock** (or press **Enter**). The same password is required every time you open the app.

Once unlocked, the **Search** tab opens and any previously indexed images are immediately available.

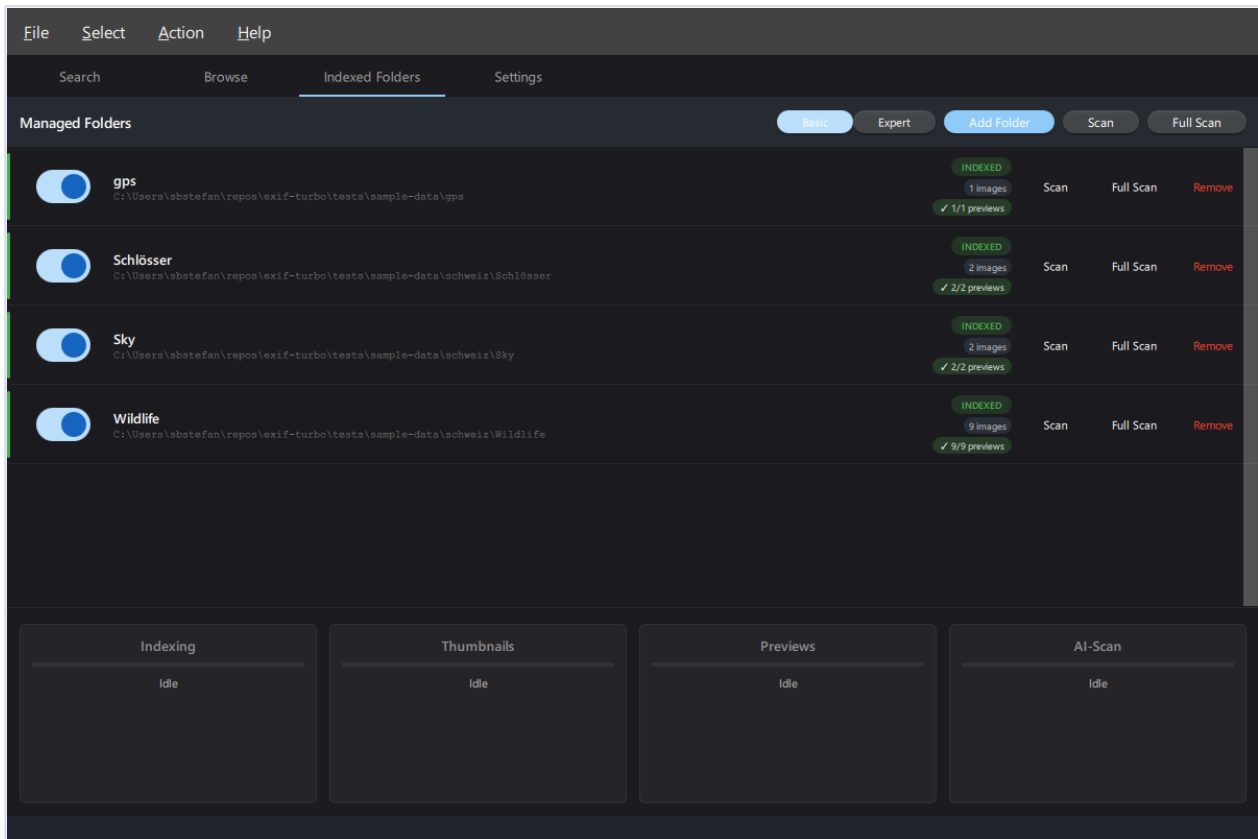
ExifTool not found: If ExifTool is not installed or not on your PATH at the time of unlock, a "ExifTool not found" dialog appears automatically. It explains that indexing is disabled and provides a link to exiftool.org. Install ExifTool and restart the app to enable indexing. You can still search and browse existing index data while ExifTool is missing.

Help menu

The **Help** menu in the menu bar provides access to this user manual, the third-party open-source licence list, and the **About** dialog, which shows the application version, a brief description of the application, and the licence (MIT).

4. Indexed Folders — Managing Your Library

Click the **Indexed Folders** tab to manage which directories are scanned.



Screenshot contains no third-party photographs.

Adding a folder

1. Click **Add Folder** in the **Managed Folders** header bar at the top of the tab.
2. Pick the directory in the file browser dialog.
3. The folder is immediately queued for scanning — its status changes to **QUEUED** then **SCANNING** once the worker starts.

The header bar provides a **Basic / Expert** mode selector. **Basic** is the default and keeps each folder row focused on the two complete workflows most people need. **Expert** exposes each operation separately. The selected mode applies to the Indexed Folders view only.

The header **Scan** and **Full Scan** buttons run the corresponding complete workflow for every enabled folder, one folder at a time. The row buttons with the same labels apply only to that folder.

Starting an index scan

In Basic mode, click **Scan** next to a folder to incrementally re-index it, refresh its sidecar tags, build missing previews, and run an incremental AI scan when AI features are enabled. The folder status changes to **SCANNING** and the progress panel appears in the bottom-right corner of the tab.

Use **Full Scan** to force every file to be re-processed, refresh sidecar tags, clear and rebuild previews, and rebuild AI data when enabled. This is useful after updating ExifTool or if you suspect any derived data is out of date.

When rescanning a single folder, only the records belonging to that folder are updated or removed. Images indexed from other folders are not affected.

Folder statuses

Each folder row shows a coloured status badge, an image-count badge (e.g. "42 images") once it has been indexed, and — if previews have been rendered — a preview-cache badge ("✓ 42/42 previews"). The preview badge is **green** when every image in the folder has a cached preview and **amber** when only some do; hover it to see the exact ratio.

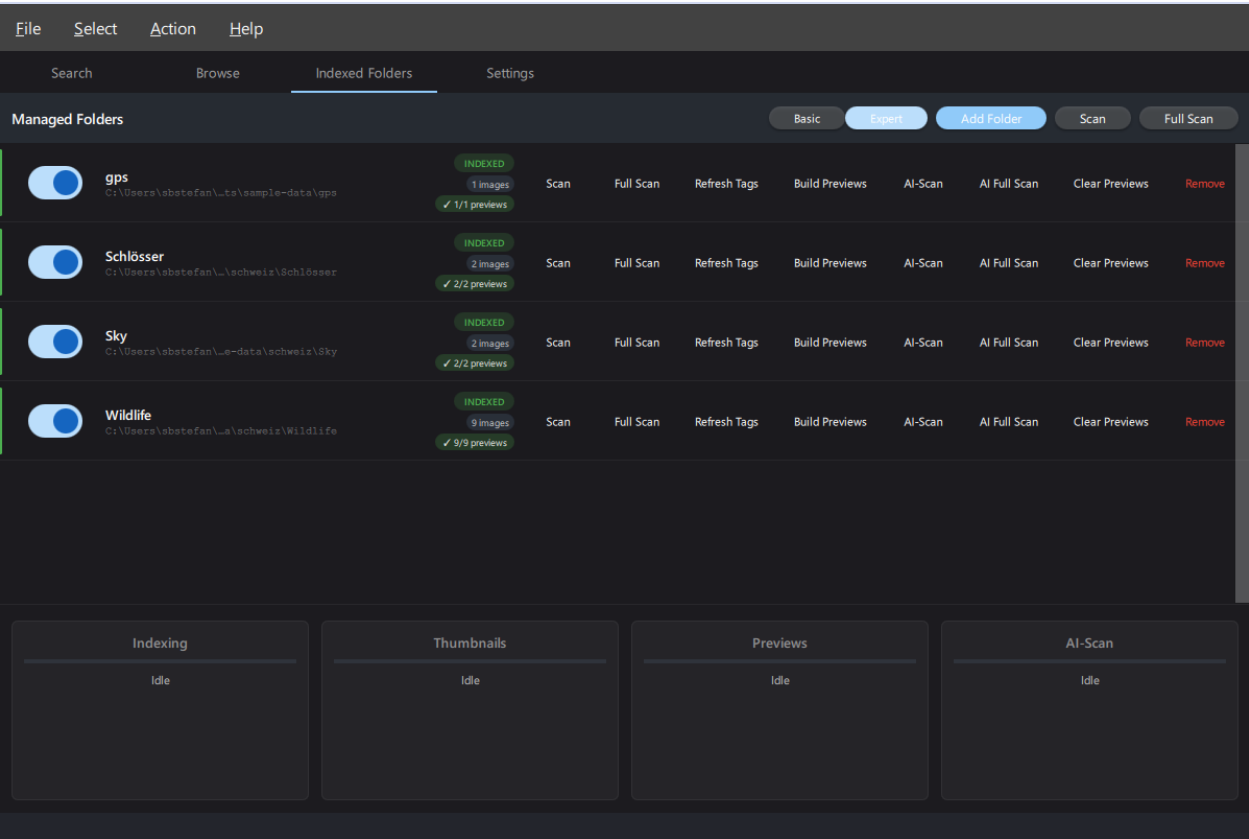
Status	Meaning
NEW	Added but never scanned
QUEUED	Waiting in the scan queue
SCANNING	Currently being indexed
INDEXED	Last scan completed successfully
MISSING	Folder path no longer exists on disk
ERROR	Last scan ended with an error
DISABLED	Excluded from search results

Disabling / enabling a folder

Toggle the **Enabled** switch to exclude a folder from search results without deleting it or its index data. Hovering the switch shows the tooltip *"Folder is included in search results"* (when on) or *"Folder is excluded from search results"* (when off).

Expert actions

Switch to **Expert** to expose the individual operations on each row:



Screenshot contains no third-party photographs.

Button	Action
Scan	Incrementally re-index this folder (only files whose modification time changed).
Full Scan	Force re-extract EXIF for every file in this folder.
Refresh Tags	Force re-read sidecar tag files for this folder's indexed images without re-extracting EXIF or rebuilding previews. Newly added and changed sidecars update the tag/search cache; deleted sidecars remove their sidecar-managed tags from the cache. Malformed sidecars are left untouched and reported as errors. The operation appears in the persistent activity panel and can be canceled there.

Build Previews	Render preview-cache JPEGs for every image in this folder. While the build is running on this folder the same button reads Cancel Previews . Disabled while another folder's preview build is in progress.
AI-Scan	Build missing CLIP vector embeddings for this folder only (incremental semantic-index build). While running, the same button reads Cancel AI-Scan . Visible only when AI features are enabled in Settings.
AI Full Scan	Rebuild every CLIP vector embedding for this folder from scratch. While running, the same button reads Cancel AI Full Scan . Visible only when AI features are enabled.
Clear Previews	Delete all cached previews for this folder. Hidden (kept invisible for layout alignment) when nothing is cached. A confirmation dialog asks <i>"Delete N cached preview(s) for \"\"? Thumbnails are unaffected."</i>
Remove	Remove the folder and delete all its indexed images. A Remove Folder confirmation dialog asks before deletion. The original files on disk are not touched. The removal then runs on a background worker behind the modal bulk-op progress overlay (see section 9), which reports each sub-step — <i>"Clearing preview cache..."</i> (cancelable, with an X / Y count) followed by <i>"Deleting index entries..."</i> .

5. Indexing Progress

The bottom of the **Indexed Folders** tab contains five persistent activity columns: **Indexing**, **Thumbnails**, **Refresh Tags**, **Previews**, and **AI-Scan**. Each column shows its own operation independently. When a column is idle, its bar is empty and its status reads **Idle**.

- **"Indexing folder N of M" / "Indexing"** — file-indexing phase. Shown while the indexer is processing folders from the queue.
- **"Thumbnails"** — thumbnail-cache phase, started automatically after indexing finishes.

- **"Refresh Tags"** — sidecar-tag refresh phase, started directly in Expert mode or after indexing in a Basic workflow. It runs in this activity column without opening the modal bulk-operation overlay.
- **"Previews"** — preview-cache phase, started by the **Build Previews** button on a folder row.
- **"AI-Scan"** — CLIP image-vector phase, started directly in Expert mode or as the final phase of a Basic workflow.

Each active column contains:

- **Progress bar** — indeterminate while the total is still being computed, then a percentage once it is known.
- **Count label** — **"Scanning for images..."** at the start of indexing discovery; **"N indexed, scanning..."** while indexing has started but the total file count is not yet known; **"Preparing..."** for the thumbnail and preview phases before the total is known; then `n / total files` (indexing) or `n / total images` (thumbnails and previews).
- **Current file** — name of the file being processed.
- **Cancel button** — stops that operation. While an indexing or thumbnail cancel is in flight the label changes to **"Canceling..."** and the button is disabled until the worker has stopped.

While AI scanning is active, the affected folder row also shows a spinner and an **AI-Scan** or **AI Full Scan** badge. Its normal folder status returns when the operation stops.

Across **all tabs**, folder-operation messages in the **status bar** begin with the indexed folder name. While indexing, refreshing tags, building thumbnails or previews, or running AI-Scan, that folder name pulses in the accent colour so the active folder remains visible even when you are working in Search or Browse. There is no separate activity dot or redundant **Indexing...** label. The status bar also shows brief event messages to its right (such as "Indexed 42 images" after a scan completes).

Self-healing cache cleanup

At the end of every successful folder index run, exif-turbo runs a quick garbage-collection pass over the thumbnail and preview cache directories. Any cached file whose source image is no longer present in the index (because the file was deleted, moved, or renamed since the last scan) is removed. The status bar shows **"Cleaning up cache..."** while the sweep runs. This keeps the cache compact and self-heals across crashes or external file deletions, so you never need to manually clear the cache directory.

Pause and resume

If you close the application while indexing is in progress, the current folder is saved as **QUEUED**. The next time you open exif-turbo and unlock the database the scan queue is automatically restored and indexing resumes where it left off.

6. Searching

The **Search** tab is the main way to find photos:

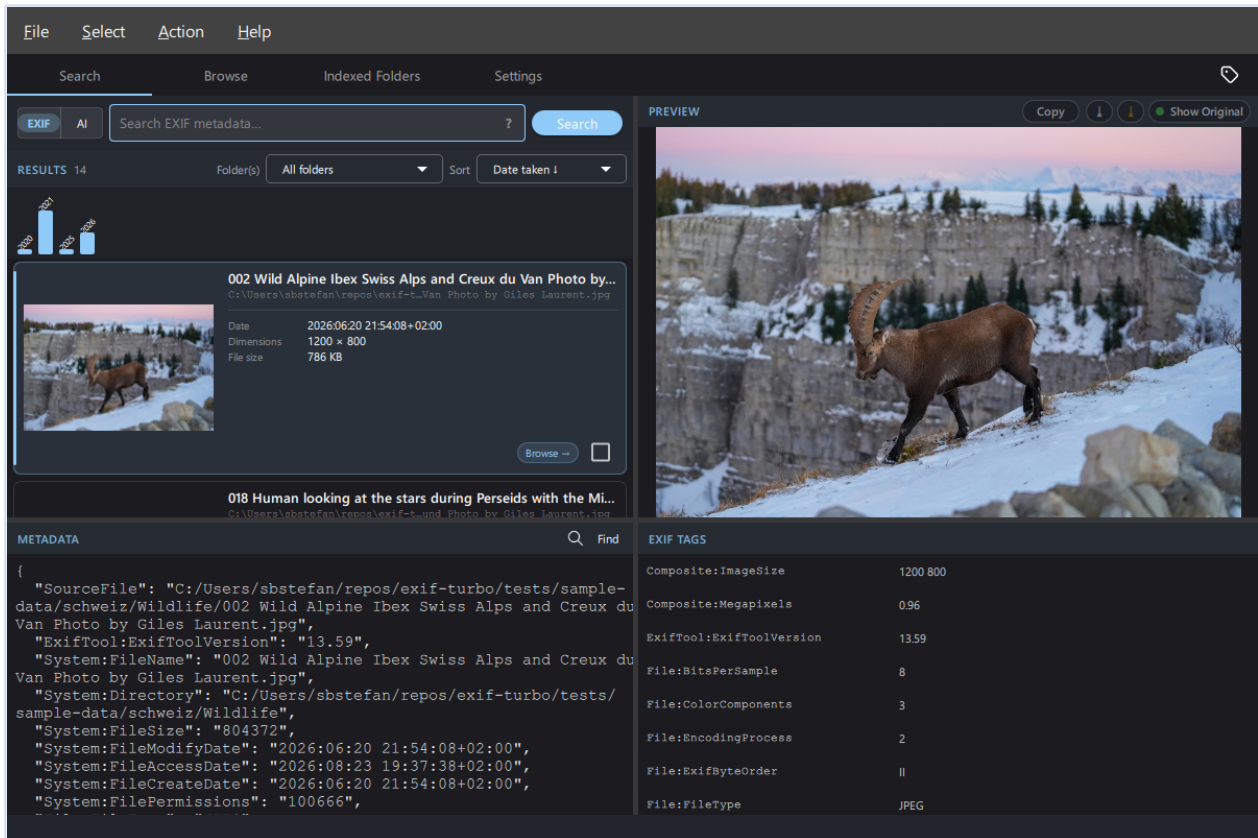


Photo: [Wild Alpine Ibex](#), © [Giles Laurent](#), License [CC BY-SA 4.0](#). Scaled/cropped in the UI; screenshot composite licensed [CC BY-SA 4.0](#).

Running a search

Type any word or phrase into the **Search EXIF metadata...** bar and press **Enter** (or click **Search**). exif-turbo performs a full-text search across every metadata field — camera make and model, lens name, date, GPS coordinates, keywords, copyright, and more.

Press **Enter** with an empty search bar to show all indexed images.

A **?** button at the right edge of the search field shows a syntax cheat-sheet when hovered. It lists six examples (single token, implicit AND, OR, NOT, exact phrase, prefix wildcard) and a short tips section. You do not need to click it — the tooltip appears automatically on hover.

When the search field contains text a **x** button appears to its left. Clicking it clears the field and immediately shows all images (equivalent to pressing **Enter** with an empty bar).

While a search is running, a semi-transparent grey overlay dims the entire UI and the cursor changes to a busy indicator. The overlay clears automatically when results are ready.

AI semantic search (EXIF/AI toggle)

When **AI Features** are enabled in **Settings**, the search bar shows an **EXIF / AI** toggle. Switch to **AI** mode to search by meaning instead of exact metadata terms.

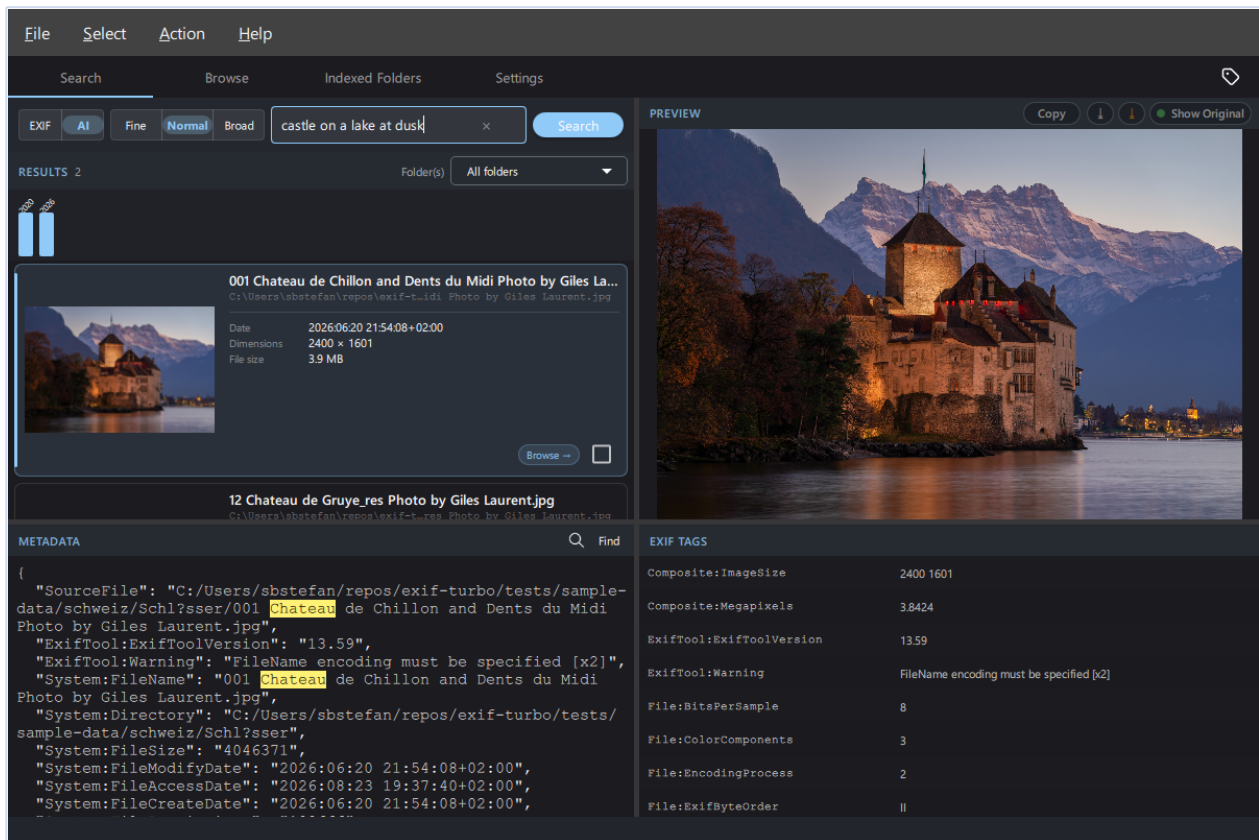


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In AI mode:

1. Enter a natural-language query such as *"golden eagle over mountain lake"*.
2. Press **Enter** (or click **Search**) to run CLIP semantic retrieval.
3. Choose a precision level:
4. **Fine**: strictest matches (score ≥ 0.22)
5. **Normal**: balanced default (score ≥ 0.20)
6. **Broad**: most permissive (score ≥ 0.18)

AI search requires CLIP vectors to exist for the target images. Build them with **AI-Scan** (or **AI Full Scan**) in the **Indexed Folders** tab.

AI vector indexes are tied to the model that created them. If the application reports missing or incompatible AI index metadata after an upgrade, run **AI Full Scan** before using AI search or tag proposals.

Note for macOS Intel users: AI features are unavailable on macOS Intel (x86_64) targets and are shown disabled in **Settings**. This is due to PyTorch support for Python 3.13+ on that platform.

Filtering by format

When results contain more than one file format, a row of format chips appears below the search bar. Click a chip to show only that format:

All	CR2 · 1459	JPG · 563	TIF · 113	PNG · 2
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Click **All** to return to unfiltered results.

Filtering by folder

When you have two or more indexed folders enabled, a **Folder(s)** dropdown appears in the RESULTS header bar, to the left of the **Sort** control. It is hidden when only one folder is indexed.

The button label reflects the current filter state:

Label	Meaning
All folders	No folder filter — results span all enabled folders
Folder name	Exactly one folder is selected
N folders	N individual folders are selected

Click **Folder(s)** to open the selection popup. Tick **All folders** to clear any active filter; tick one or more folder names to restrict results to those folders only. Multiple folders can be selected simultaneously. Hovering over a folder name shows its full path as a tooltip.

Drive roots on Windows — when an entire drive (e.g. C:\) is added as an indexed folder it appears in the dropdown with a friendly label such as **"OS (C:)"** (volume name + drive letter), rather than a blank entry.

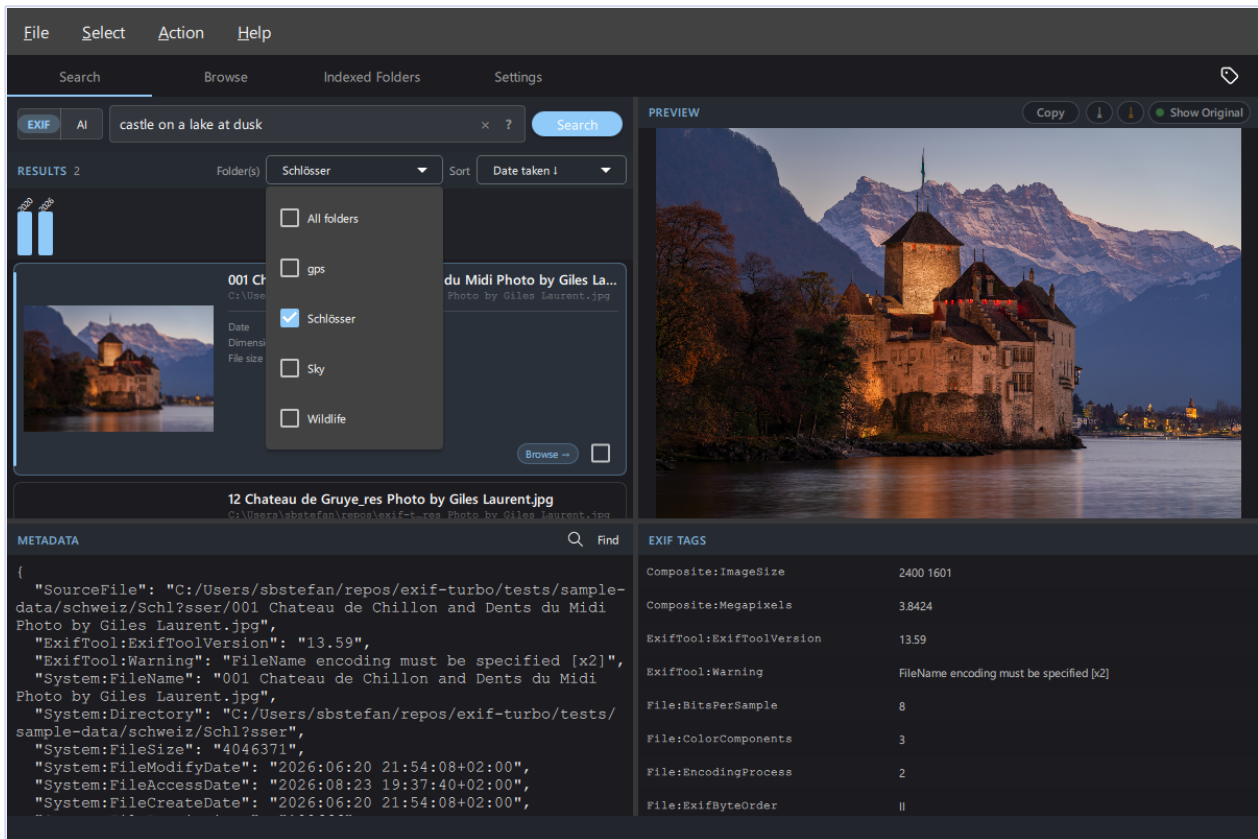


Photo: [Chateau de Chillon and Dents du Midi](#), © [Giles Laurent](#), License [CC BY-SA 4.0](#).

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The selection chip

Whenever at least one image is marked (or the marked-only filter is active), a **selection chip** appears in the RESULTS header bar next to the **Folder(s)** dropdown. It shows the current marked count:

Label	Meaning
■ N	Every marked image is in the current results
■ here/total	Only some marked images match the current filters (e.g. ■ 12/87)

Click the chip to toggle the **marked-only filter** — the tooltip switches between *"Show only selected images"* and *"Show all results"*. While the filter is active the chip stays highlighted so you always know which view you are looking at.

Sorting results

Use the **Sort** dropdown at the top-right of the results panel:

Option	Description
Date taken ↓	EXIF capture date, newest first (default)
Date taken ↑	EXIF capture date, oldest first
Name A→Z	Filename ascending
Name Z→A	Filename descending
Path A→Z	Full path ascending
Path Z→A	Full path descending
Largest	File size, largest first
Smallest	File size, smallest first

Date taken ↓/↑ sorts by the capture timestamp stored during indexing. exif-turbo resolves this timestamp from a prioritised chain: primary EXIF fields (`DateTimeOriginal`, `CreateDate`) are tried first; if none are present, XMP, IPTC, and QuickTime creation fields are consulted; as a last resort the file-system creation time (macOS/Windows) or modification time (Linux) is used. Infrastructure metadata such as ICC colour-profile dates is never used. Images with no resolvable date appear at the end of the list in both directions.

The chosen sort order is remembered per database and restored automatically the next time you open the application.

Filtering by capture year

Whenever at least one indexed image has a known capture date a **year histogram** appears below the format chips in the Search tab. Each bar represents one calendar year; its height is proportional to the number of matching images taken in that year relative to the busiest year. A small spinner appears in this area while the timeline counts are loading or refreshing.

Action	Effect
Click a bar	Filter results to images captured in that year
Shift-click a different bar	Extend the selection to cover the range between the two bars
Click the active bar again	Clear the year filter

× chip (right side of histogram)	Clear the year filter
---	-----------------------

A tooltip on each bar shows the year and image count. When a filter is already active, hovering a bar that is not the sole selected year shows "Shift-click to extend range".

Images that have no resolvable capture date are excluded from results while a year filter is active.

Loading more results

Results are loaded in batches as you scroll. When you reach the bottom of the result list the next batch is fetched automatically. The total match count is shown next to the **RESULTS** badge in the panel header.

Opening images from results

Single-click a result card to select it and load the preview and metadata panels.

Double-click a result card to open the file or folder in your system's default application: -
Double-clicking the **thumbnail** (left side of the card) opens the image in your default image viewer. -
Double-clicking the **info area** (right side of the card) opens the parent folder in the system file manager. On Windows, Explorer opens with the file highlighted; on macOS and Linux the parent folder is opened.

If the original file cannot be reached — for example when the folder that contains it is not currently mounted or attached — nothing is opened and a **red warning** appears in the status bar at the bottom of the window naming the indexed folder you need to (re)attach or mount, e.g. *"Original data source not attached: D:\Photos — attach or mount this folder to open files."*

Search examples

Query	What it finds
Canon EOS R5	All images shot with a Canon EOS R5
f/1.4	All images taken at f/1.4 aperture
Switzerland 2024	Images with Switzerland and 2024 in any metadata field
eagle	Images whose filename, title, or keywords mention eagle
Nikon Z 9 ISO 6400	Nikon Z 9 shots at ISO 6400

Advanced query syntax

The search box accepts the full **SQLite FTS5** query language:

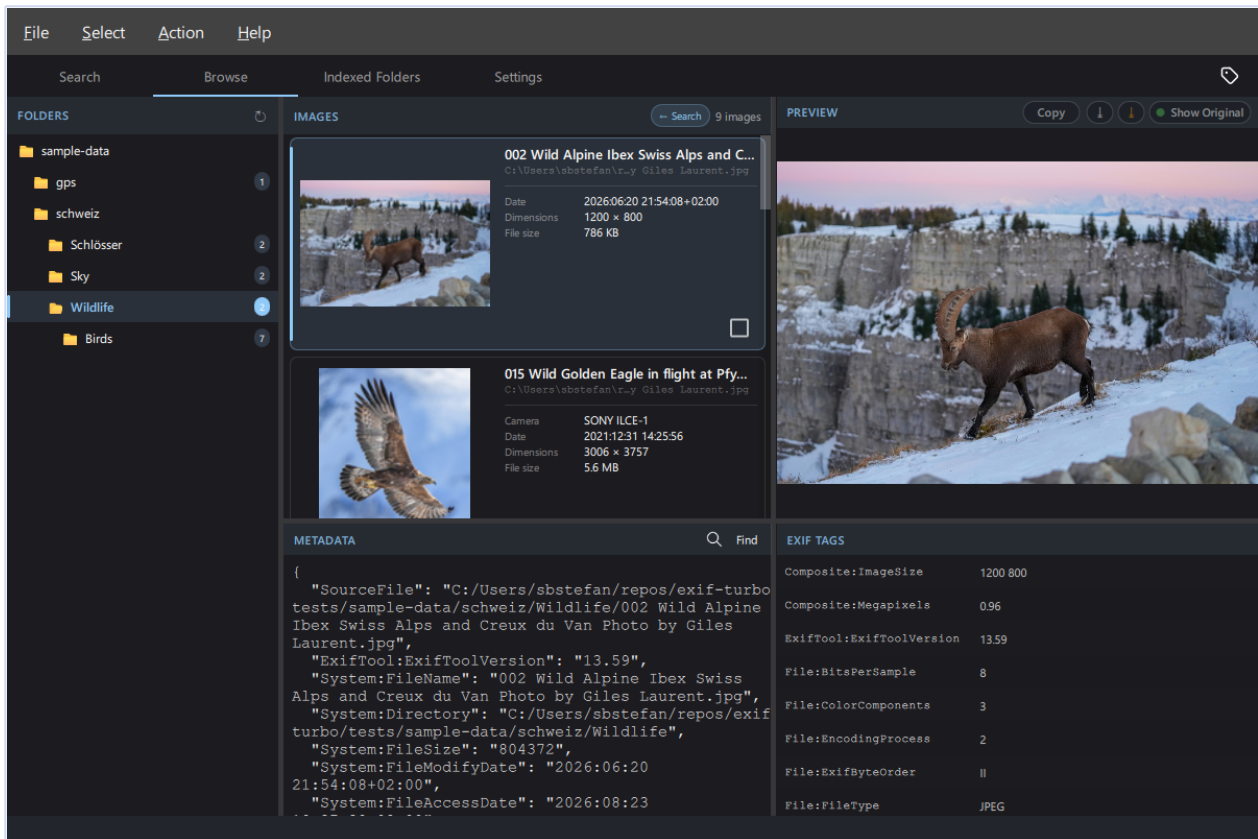
Syntax	Example	What it does
term	Canon	Keyword anywhere in metadata
"exact phrase"	"red deer"	Terms must appear adjacent and in order
term1 AND term2	Canon AND 50mm	Both terms must be present
term1 OR term2	Canon OR Nikon	Either term must be present
term1 NOT term2	50mm NOT Nikon	First term present, second term absent
prefix*	Fuji*	Matches any token starting with the prefix

Multiple terms without an operator (Canon 50mm) are treated as an implicit AND.

ExifTool stores metadata with group-prefixed keys such as `GPS:GPSLatitude` or `ExifIFD:FocalLength`. You can type these directly in the search box — the colon is treated as a word separator, so `GPS:GPSLatitude` becomes an implicit AND of `GPS` and `GPSLatitude`.

7. Browsing by Folder

The **Browse** tab lets you navigate your library by folder hierarchy:



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The left panel shows all indexed folders as an indented list — sub-folders are indented under their parent. Each entry shows the folder name and a count of images inside it. Click any folder to show only its images in the centre panel. Click the highlighted folder again to deselect it and show all images.

The same thumbnail list and preview pane appear as in Search. **Single-click** an image to select it and load the preview. The preview supports the same zoom and pan gestures as in Search (scroll wheel, touchpad scroll, touchpad pinch-to-zoom, drag-to-pan, double-tap to reset — see [Zooming and panning](#)). **Double-click** an image to open it in your system's default viewer. (Unlike the Search tab, double-clicking in Browse always opens the image — there is no folder-open shortcut.)

The Browse tab shows the same **METADATA** and **EXIF TAGS** panels as the Search tab, including the GPS location bar. They appear in a split view below the image list and preview panel. The inline **Ctrl+F** find bar works on the Browse tab too — click **Find** in the METADATA panel header (or press **Ctrl+F** when the Browse tab is active) to search within the metadata text.

The Browse tab preview pane has the same toolbar pill buttons as the Search tab:

- **Copy** — copies the currently displayed preview image to the system clipboard. A brief toast notification confirms the action.
- **Save Preview As** (white ■) — opens a **Save File** dialog to save the preview as JPEG or PNG.

- **Save Original As** (orange ■) — copies the original source file byte-for-byte to a destination you choose.
- **Show Original / Show Preview** toggle — switches between the cached preview (green dot) and the full-resolution source file (orange dot). Only shown when the selected image has a cached preview.

While a full-resolution original is loading, a "**Loading original...**" spinner overlay appears over the preview area (only when **Show Original** is active).

Navigating the Browse image list

The image list takes keyboard focus as soon as a folder is selected. You can move through images without touching the mouse:

Key	Action
↓	Select the next image
↑	Select the previous image
Page Down	Jump one page forward
Page Up	Jump one page backward

A permanently visible vertical scrollbar (12 px) is always shown on the right edge of the list so you can drag it or click to jump to any position without first hovering the right edge.

Search-tab filter state is preserved

The Browse tab has its own independent navigation context. When you switch from Search to Browse and back, your active Search query, format chip, date range, sort order, extension filter, and folder filters are all restored exactly as you left them — Browse navigation never affects the Search state.

Switching back to the **Search** tab restores the search state exactly as you left it. Any image previously selected while browsing is not automatically highlighted in Search.

Jumping between Search and Browse

Each result card in the **Search** tab has a **Browse** → pill button in its bottom-right corner (to the left of the selection checkbox). Clicking it:

1. Switches to the **Browse** tab.
2. Navigates to the folder that contains that image.

3. Scrolls the Browse image list to that exact image and selects it.

The Browse tab image list header has a ← **Search** back button. Clicking it returns to the **Search** tab and restores the exact scroll position and previously-selected image — no need to re-enter the search query.

8. Viewing Metadata and EXIF Tags

Selecting any image in the result list populates three panels:

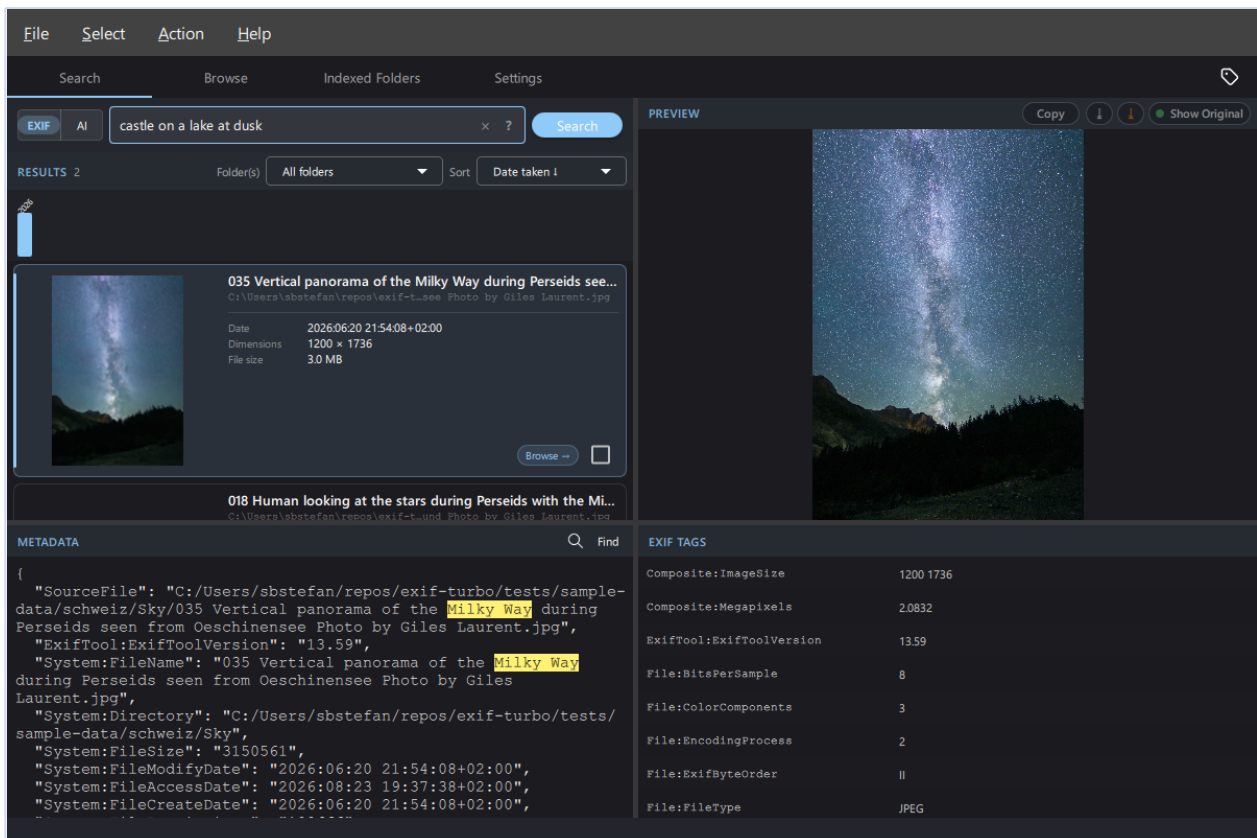


Photo: [Vertical panorama of the Milky Way during Perseids](#), © [Giles Laurent](#), License [CC BY-SA 4.0](#).
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Preview

The right pane shows a full-resolution preview of the selected image scaled to fit the available space. For RAW files (CR2, CR3, NEF, ARW, DNG, etc.) the embedded preview JPEG is used. A cached thumbnail is shown immediately as a low-resolution placeholder while the full image is loading; the full image fades in once it has been decoded.

Show Preview / Show Original

When the selected image has a cached preview JPEG, a small pill toggle appears at the right edge of the **PREVIEW** header bar. The label tells you what clicking it will do:

- **Show Original** (with a green dot) — you are currently looking at the cached preview; click to load the full-resolution source file.
- **Show Preview** (with an orange dot) — you are currently looking at the full-resolution source; click to switch back to the cached preview.

The pill is useful for RAW files: the cached preview decodes instantly and is sufficient for most viewing, but switching to **Show Original** loads the full-resolution image when you want to zoom in for detail. The toggle is hidden when no cached preview is available for the selected image.

While a full-resolution original is loading, a small "**Loading original...**" overlay with a spinner appears over the preview area so you know the image is being decoded (large RAW files may take a few seconds).

The **Show Preview / Show Original** toggle is available in both the **Search** and **Browse** tabs.

Copying the preview image

You can copy the currently displayed preview image to the system clipboard in two ways:

- Click the **Copy** pill button in the **PREVIEW** header bar toolbar (the leftmost of the pill buttons).
- **Right-click** anywhere on the preview image and choose **Copy Image to Clipboard** from the context menu.

Both actions are available in the **Search** and **Browse** tabs. A brief toast notification at the bottom of the window confirms that the image has been placed on the clipboard. If rendering fails for any reason, the file path is copied as plain text instead.

Saving the preview image

A **Save Preview As** button (white ■) appears in the **PREVIEW** header bar toolbar — the second pill from the left — when an image is selected. Clicking it opens a native **Save File** dialog with the filename pre-filled as `<stem>_preview.jpg`, where `<stem>` is the original file's base name without its extension. The available save formats are:

- **JPEG** (.jpg / .jpeg)
- **PNG** (.png)

The saved image is the currently displayed preview. If the **Show Original** toggle is active, the high-resolution source is used; otherwise the cached preview JPEG is saved.

A brief toast notification "**Preview saved**" confirms the export.

You can also trigger this action by right-clicking anywhere on the preview image and choosing **Save Preview As...** from the context menu.

Saving the original file

A **Save Original As** button (orange ) appears in the **PREVIEW** header bar toolbar — the third pill from the left — when an image is selected. Clicking it opens a native **Save File** dialog with the original filename pre-filled as the suggested name. The source file is copied byte-for-byte to the chosen destination — no re-encoding takes place.

A brief toast notification "**Original saved**" confirms the copy.

You can also trigger this action by right-clicking anywhere on the preview image and choosing **Save Original As...** from the context menu.

Recreate Thumbnail / Recreate Preview

The preview context menu (right-click on the preview image) also offers two rebuild actions, useful when a cached thumbnail or preview ever looks wrong (for example a video frame that was extracted before the rotation fix):

- **Recreate Thumbnail** — deletes the cached thumbnail for the selected image (including any `.skip` sentinel that previously marked it as unthumbnaible) and re-queues thumbnail generation. The thumbnail in the left-hand result grid refreshes automatically as soon as the new file is written — you do not need to switch images or restart the app.
- **Recreate Preview** — deletes the cached preview JPEG for the selected image. The preview pane re-renders the image immediately on next display.

Both actions only affect the selected image; other cached thumbnails and previews are left untouched.

Zooming and panning

The preview supports cursor-anchored zoom and drag-to-pan:

Input	Action
Ctrl + scroll wheel (mouse)	Zoom in / out, anchored to the cursor position
Ctrl + two-finger scroll (touchpad)	Zoom in / out, anchored to the cursor position
Touchpad — pinch	Zoom in / out, anchored to the pinch centroid
Click and drag / two-finger drag	Pan the zoomed image in any direction
Plain scroll wheel / two-finger scroll	Pan vertically when zoomed in

Double-click / double-tap	Reset zoom to fit
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Zoom is capped at **16×**. A badge in the bottom-right corner of the preview shows the current zoom level when it is above 1×. Selecting a new image always resets the zoom to fit.

Metadata panel (bottom-left)

Displays the metadata for the selected image as formatted, indented JSON. Click the **Find** button in the panel header (or press **Ctrl+F**) to open an inline search bar and find any tag value. Press **F3** (or click the ▼ / ▲ arrows) to jump through all matches. Search terms from the main search bar are highlighted automatically.

GPS location bar

When the selected image contains GPS coordinates in its EXIF metadata, a thin bar appears directly below the **Find** bar. It shows a ■ icon, a **"GPS location —"** label, and three links you can click to view the location on a map:

Link	Service
OpenStreetMap	Opens the coordinates on OpenStreetMap at zoom level 14
Google Maps	Opens the coordinates in Google Maps
GeoHack	Opens the Wikimedia GeoHack coordinate hub — lists nearby Wikipedia articles and dozens of map services

The bar is hidden when the selected image has no GPS data, or when nothing is selected. Hovering over a link shows the full URL in a tooltip.

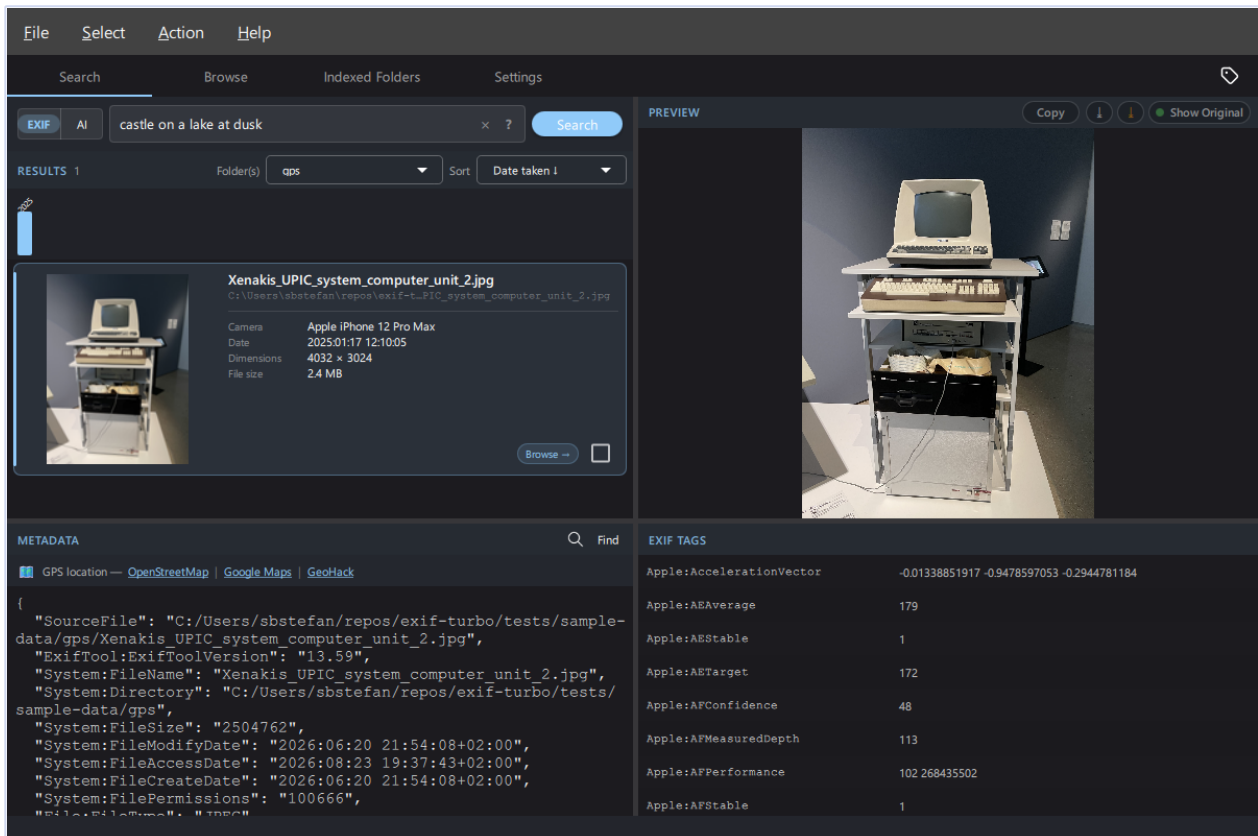


Photo: [Xenakis UPIC system computer unit](#) by 1904.CC (Manuel Schmalstieg), [CC0 1.0](#), via [Wikimedia Commons](#). Scaled/cropped in the UI; attribution is voluntary.

EXIF Tags panel (bottom-right)

Displays the same metadata as a clean two-column list, sorted alphabetically. Hover over a truncated tag or value to see the full text in a tooltip.

You can drag the divider between the two bottom panels to adjust the split. By default they start at 50 % / 50 %.

9. Marking Images & Bulk Actions

Every result card has a **checkbox** in its bottom-right corner. Tick it to *mark* that image — marks persist across searches, tab switches, and app restarts (the state is stored in the encrypted database). The same checkbox is present on the cards in the **Browse** tab, so you can mark images while browsing as well as while searching. Marked images are the input for the bulk actions in the **Action** menu.

The menu bar exposes two menus dedicated to marking and bulk actions.

Select menu

Action	What it does
Select All	Marks every image matching the current filters (search query, format chip, folder filter, marked-only toggle). Runs on a background thread in batches so the UI stays responsive.
Deselect All	Unmarks every image matching the current filters.
Invert Selection	Flips the marked state of every image matching the current filters.
Select Images Without Thumbnail	Marks every matching image whose thumbnail has not been cached on disk yet. Images the thumbnailer has permanently given up on (oversized files or decoder errors, recorded as a <code>.skip</code> sentinel) are also marked, so you can act on them — for example by exporting the metadata or deleting the offending files. Like the other Select actions, it respects the active search query, format chip, folder filter, and marked-only toggle.

All four actions show the **bulk-op progress overlay** with a live `X / Y` counter and a **Cancel** button.

The same modal overlay is also reused for the two long-running maintenance actions — **Remove Folder** (Indexed Folders tab) and **Reset Database** (Settings tab). For those, the overlay adds a sub-step detail line describing the current phase, and for phases that must not be interrupted (such as the database vacuum during a reset) it hides the **Cancel** button and shows a *"This step cannot be canceled..."* notice instead.

Action menu

The Action menu operates on whatever is currently marked across the entire database — not just the visible results.

Action	What it does
--------	--------------

Export Metadata as JSON...	Writes the EXIF metadata of every marked image to a JSON file you choose via a Save dialog. The export honours the current Sort order (date taken, filename, path, or size). When nothing is marked, the menu label changes to <i>"Export Metadata as JSON... (all results)"</i> and the action exports every image matching the current filters instead. The on-disk layout is controlled by Settings → JSON Export Formatting (see <i>Settings</i>).
Generate Tagged Derivatives for Current Results...	Asks for a target folder, then generates derivatives for every image matching the current search and filters, including result pages that have not been loaded into the view.
Generate Tagged Derivatives for Marked Images...	Asks for a target folder, then generates derivatives for all marked images in enabled indexed folders. Disabled when nothing is marked.
Delete Marked Images...	Permanently deletes every marked image from disk and removes its row from the index. Cached thumbnails (<code>.png</code> / <code>.enc</code>), <code>.skip</code> sentinels, and any rendered preview (<code>.jpg</code> / <code>.jpg.enc</code>) for the deleted images are also cleaned up. Disabled when nothing is marked. The menu label includes the current count, e.g. <i>"Delete Marked Images... (12 selected)"</i> .

Long-running actions show progress and support cancellation. Any completed derivatives remain in the target folder, and any deletions already made stay on disk and in the index; the database is never out of sync with the file system.

Confirming a delete

Because **Delete Marked Images...** is irreversible, the confirmation dialog requires an extra step:

1. The dialog states how many files will be deleted (e.g. *"Permanently delete 12 marked image file(s) from disk and remove them from the index?"*).
2. A text field asks you to **type the exact count** — for example `12` — into the box labelled *"To confirm, type the number 12 below"*.
3. The **Yes** button stays disabled until the typed number matches the expected count exactly. Pressing **Enter** in the field also confirms the action when the count matches.

4. **Cancel** (or closing the dialog) aborts without deleting anything.

After completion the status bar reports the outcome, e.g. **"Deleted 12 image(s)."** When some files were already gone or could not be removed (for example because of file-system permissions) extra clauses are appended: **"3 were already missing."** and/or **"1 could not be deleted."**

There is no undo. Files are removed using the operating system's regular delete call — they do *not* go to the Recycle Bin / Trash. Make sure the marked set is correct before confirming.

Tip — auto-sized menus

The Select and Action menu popups are sized to fit their longest item, so dynamic labels such as *"Delete Marked Images... (1234 selected)"* are always shown in full and never truncated.

10. Tagging with Wikidata

Tagging uses a bundled, curated 8,339-concept visual vocabulary from Wikidata without writing metadata into the original image. Its reviewed 8,200-concept base is extended with qualified concepts linked to the Library of Congress TGM. It is disabled by default for each database; enable it under **Settings** → **Tagging and Controlled Vocabulary**. The snapshot is available fully offline and requires no vocabulary or translation-pack installation. It is a controlled visual vocabulary, not an exhaustive copy of Wikidata.

Every bundled concept has intrinsic preferred labels and aliases for exactly English, German, French, and Italian. The snapshot records its source dump and manifest checksums and is distributed under Wikidata's CC0 terms. New accepted QIDs are written in sidecar schema v2. Existing schema-v1 `loc-tgm` tags remain readable and retain their canonical-label and old localization-overlay fallback for display and export; the legacy TGM importer and parsers remain maintenance tooling, not a user-facing download workflow.

Sidecars and search

The first tagging change for `photo.jpg`, including an embedded-tag exclusion, creates `photo.jpg.sidecar.json` beside the image. Sidecars are deterministic UTF-8 JSON and are the authoritative store for accepted tags and derivative exclusions. They are plain text: SQLCipher database encryption does **not** encrypt them, so they inherit the source folder's permissions and backup policy. Tagging never changes the original image's bytes or timestamp.

Each accepted Wikidata term stores a qualified ID such as `wikidata:Q4421`, its canonical English label, subject or genre/form category, and acceptance provenance in schema v2. Custom tags are stored separately in the same sidecar as normalized text labels. Legacy `loc-tgm:tgmNNNNNN`

records are preserved.

Accepted canonical labels, qualified IDs, categories, vocabulary identity, all bundled localized preferred labels/aliases, legacy aliases when available, and custom labels are copied into the encrypted database's FTS5 cache. Search uses the normal EXIF query box and syntax; undecided and rejected proposals are not searchable. A regular or full image scan synchronizes new, changed, or deleted sidecars even when the original image stamp did not change. Malformed sidecars are reported and left untouched.

Move or rename a sidecar together with its image. Version 1 does not infer an external rename. Removing an image or indexed folder clears only database cache rows; it does not delete a sidecar. Deleting a sidecar removes its accepted tags from FTS after the next synchronization.

Tagging drawer

From the **Search** or **Browse** tab, click the tag button at the upper right or press **Ctrl+T**. The non-modal drawer contains these controls:

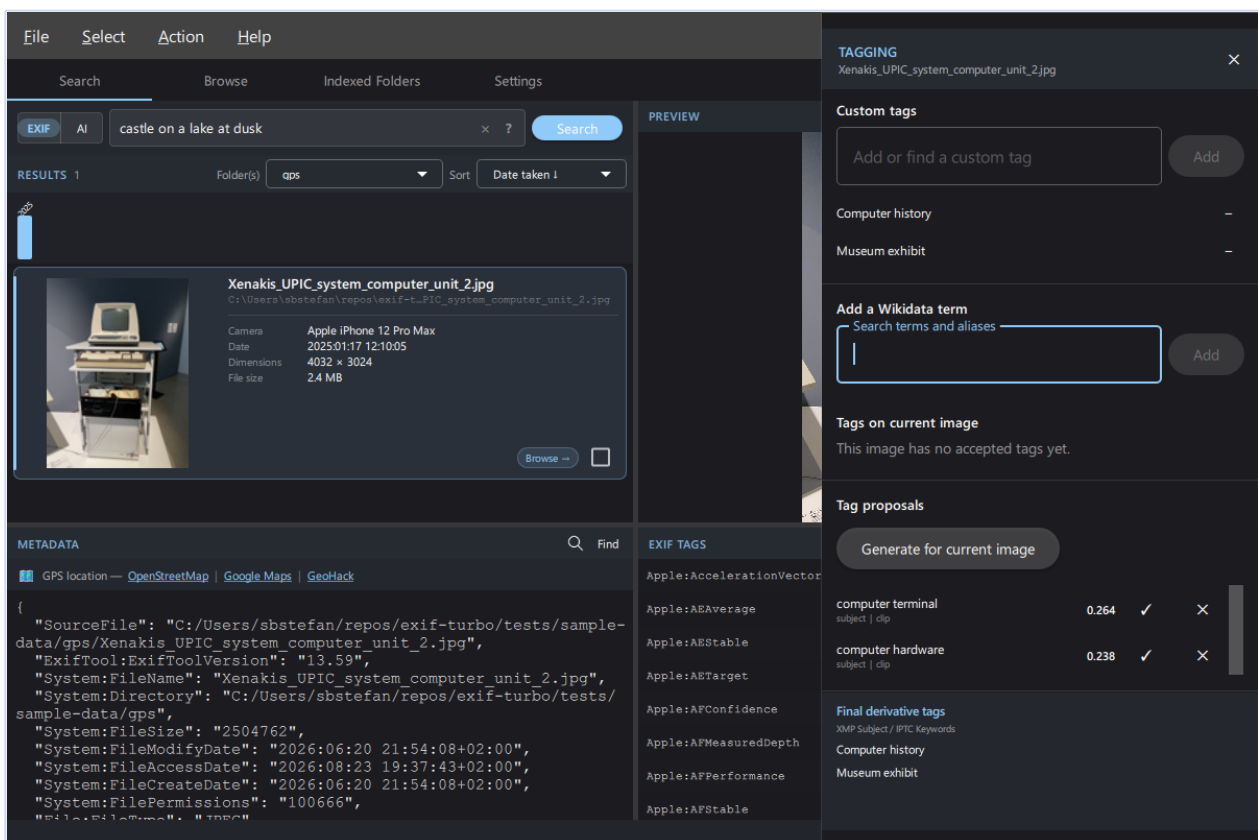


Photo: [Xenakis UPIC system computer unit](#) by 1904.CC (Manuel Schmalstieg), [CC0 1.0](#), via [Wikimedia Commons](#). Scaled/cropped in the UI; attribution is voluntary.

- **Existing image tags** lists keywords already embedded in the original image's XMP or IPTC metadata. Select individual tags to exclude them from derivatives, or select **Ignore all existing tags**. These choices are saved in the image sidecar. They never change the source image.

- **Add a Wikidata term** searches preferred labels and aliases in the selected metadata language after a short delay. Select a result and click **Add** to apply it to the focused image. **Enter** accepts the highlighted result, and **Down** moves through results.
- **Tags on current image** shows the focused image's canonical tags, category, and provenance. The minus button removes a tag from that image.
- **Custom tags** accepts a new label with **Enter** or **Add**. Previously used labels are remembered for the current database and appear as suggestions; click one to reuse the same spelling. Removing a custom tag from an image does not remove it from the remembered list. Custom tags do not require AI.
- **Tag proposals** generates suggestions automatically when the drawer opens and whenever the focused image changes. Each row shows its score and provider and has accept and reject buttons. **Generate for current image** refreshes them manually. Undecided suggestions are kept only for the current selection and are not restored after restart. Rejected proposals remain suppressed for the current Wikidata snapshot, prompt, and model fingerprint.
- **Final derivative tags** remains visible in the fixed footer. It previews the sorted, deduplicated union of included embedded keywords and accepted controlled/custom additions that a derivative will receive.
- **Copy tags to other images** copies the focused image's accepted controlled and custom tags and its embedded-tag ignore settings to a selected target set. Individual ignore entries are copied only when that embedded tag exists on the target. Choose **Marked images**, **Current search results** (the complete result set, including unloaded pages), or **Current folder** while browsing. **Add** keeps each target's existing tags and ignore settings and adds the source values. **Replace** substitutes the target's controlled tags, custom tags, and ignore settings and requires confirmation. The focused source image is always excluded. Progress, cancellation, and a completion summary are shown in the drawer.

Long-running vector and proposal operations show progress and a **Cancel** button. Derivative generation starts from the **Action** menu; the drawer footer is only a read-only preview.

Marks and bulk scopes

Direct add, remove, and proposal-review controls edit only the focused image. The drawer's **Copy Tags** action can use the persistent marked set as a bulk target, while marks also remain available as a derivative scope. Press **Space** to toggle the focused image's mark, then choose **Marked images** in Copy Tags or use **Action** → **Generate Tagged Derivatives for Marked Images...**

CLIP proposal prerequisites

Manual Wikidata search and tagging do not require AI or network access. CLIP model assets require network access on first AI use unless they are already cached; proposal generation then works offline. Proposals require:

1. **AI Features** enabled in Settings. This is unavailable on macOS Intel.
2. Image CLIP vectors built separately with **AI-Scan** or **AI Full Scan** for the relevant indexed folder.
3. A separate Wikidata term-vector index built with **Build Vectors** under **Tagging and Controlled Vocabulary**.

The image FAISS index remains image-only; Wikidata concepts are stored in a separate FAISS index. Image-vector schema v2 stores five views per image: the full image and four overlapping corner crops. Existing image indexes require **AI Full Scan** because incremental AI-Scan cannot reconstruct missing crop vectors. Term-vector schema v3 stores separate English, German, French, and Italian prompt rows for every QID. A bundled snapshot or prompt change makes term vectors stale and requires **Rebuild Vectors**, but does not require rebuilding image vectors.

The displayed score is the maximum cosine similarity across the image's five views and the QID's four locale prompts. Proposal generation never scans original images implicitly: a missing image vector is reported as requiring an AI scan.

The proposal threshold defaults to **0.20**. Scores are model- and dataset-dependent similarities, not calibrated probabilities. Proposals are always review-only and become tags only after explicit user acceptance.

Tagged derivatives

Use **Action** → **Generate Tagged Derivatives for Current Results...** to process the complete current result set, including pages not yet loaded into the view, or **Action** → **Generate Tagged Derivatives for Marked Images...** to process the marked set. Only images with accepted tags produce copies. The chosen output root must be outside every indexed source root. The exporter:

Under **Settings** → **Tagging and Controlled Vocabulary**, choose a **Metadata language** independently from the interface language. It controls Wikidata lookup, display, and interface-mode export. Canonical mode emits canonical English; metadata-language mode emits that locale's intrinsic preferred label; selected mode emits deduplicated preferred labels for selected `en`, `de`, `fr`, and `it` locales. Custom tags and preserved embedded keywords are never translated. Legacy `loc-tgm` tags retain canonical fallback and old overlay behavior.

- preserves each source format and relative source folder tree;
- adds collision-safe top-level labels when marks span multiple indexed roots;
- skips untagged images and existing destination files without overwriting;
- merges existing embedded keywords with accepted controlled and custom labels, removes case-insensitive duplicates, and writes the result to **XMP Subject** and **IPTC Keywords** on a temporary copy;
- reads the copied file immediately before writing so non-excluded keywords changed since the last index scan are preserved, then verifies both fields with ExifTool;
- removes an incomplete temporary copy after a write or verification failure;

- never copies sidecars into the derivative tree.

Originals are explicitly forbidden as metadata-write targets. Other copied metadata is preserved, but version 1 does not convert formats, provide custom metadata mappings, or overwrite existing derivatives. Some source formats may not support the requested writable metadata; those items are reported as failures and the source remains unchanged.

Lifecycle and reset

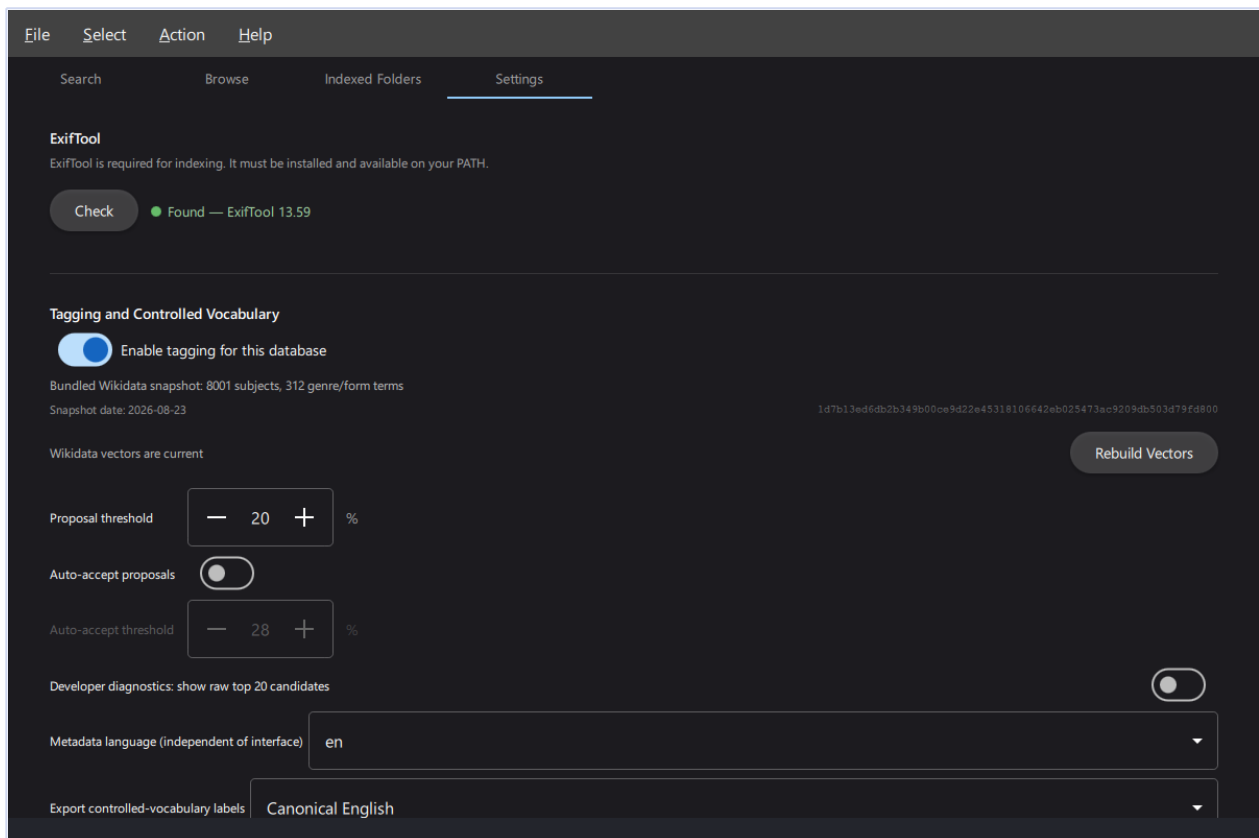
Disabling tagging hides the workbench but does not delete sidecars, rejected proposal decisions, legacy TGM data, or cached accepted tags. Already synchronized tags remain searchable. Closing the app requests cancellation of running tagging workers; completed item-level writes remain in place, while undecided suggestions are discarded.

Reset Database clears image/tag/proposal rows, marks, indexed folders, thumbnail and preview caches, and the per-database controlled-term vector index. It deliberately does not traverse source folders to delete adjacent sidecars. The separate image AI index files are not explicitly deleted by reset; use **AI Full Scan** after rebuilding the image index when a clean semantic index is required. Re-add and scan folders to synchronize sidecars, then rebuild Wikidata vectors before generating proposals.

11. Settings

Click the **Settings** tab to configure application behaviour.

Tagging and Controlled Vocabulary



Enable tagging for this database controls the drawer UI. The section shows the bundled Wikidata snapshot's subject and genre/form counts, date, and checksum. There is no vocabulary or translation-pack installer. **Build Vectors** / **Rebuild Vectors** creates the separate CLIP Wikidata term index and is enabled only when AI is available and on.

Proposal threshold defaults to 20%. **Developer diagnostics: show raw top 20 candidates** bypasses the proposal threshold for proposal generation. It displays the winning image view and prompt locale alongside each decimal cosine-similarity score. Every proposal still requires explicit acceptance.

Worker Threads

Controls the number of parallel threads used for indexing and thumbnail generation. Higher values speed up processing on multi-core machines but use more CPU and memory. The default is half the number of detected CPU threads.

Preview Cache Size

Long-edge resolution used by the preview-cache builder. Larger values give sharper detail when zooming but take more disk space and longer to render. Choose a value (in pixels) from the dropdown — the change applies to subsequently built previews; existing cached previews are unaffected until you rebuild them via **Build Previews** on a folder row.

libvips Allowed Extensions

Controls which file extensions may use the native libvips fallback for images that are unusually large or cannot be decoded correctly by Pillow. The default list covers JPEG, PNG, TIFF, WebP, and GIF. Add an extension such as `.bmp` only when you need libvips support for that format; remove an extension to prevent that format from reaching libvips. Normal Pillow decoding remains available.

Changes apply immediately and are stored per database. Operations that libvips marks as untrusted remain blocked regardless of this list; adding an extension does not weaken that mandatory protection.

Indexing Blacklist

A list of file and folder name patterns to skip during indexing. Supports wildcards: `*` matches any sequence of characters, `?` matches a single character. Patterns are matched against individual file or folder names, not full paths.

Examples: `@eaDir`, `*.tmp`, `Thumbs.db`

Changes to the blacklist take effect on the next rescan.

JSON Export Formatting

Controls how the **Export Metadata as JSON...** action writes its output.

- **Pretty-print (indented) JSON** — off by default, which keeps the historical compact layout (the export is a JSON array with each record serialised on its own line). Turn it on to indent every record for easier human reading.
- **Indent style** — when pretty-printing is on, choose **Spaces** or **Tabs**.
- **Indent size** — when the style is **Spaces**, choose how many spaces make up one indentation level (2, 4, or 8).

Whatever the format, the file is always valid JSON that round-trips back to the same records. The setting is stored per database and applies to the next export.

Theme

Choose between **system** (follows OS dark/light mode), **light**, or **dark**. The theme changes immediately.

Language

Select the application display language from the dropdown. A restart is required for that change to take full effect. This does not change controlled vocabulary lookup or metadata export language;

configure those separately under **Tagging and Controlled Vocabulary**.

ExifTool

Click **Check** to verify that ExifTool is installed and available on your `PATH`. A colour-coded badge appears next to the button:

- **Green dot + "Found — ExifTool \<version>"** — ExifTool was found and the version is shown.
- **Red dot + "Not found"** — ExifTool is missing. A download link (exiftool.org) and a restart hint are shown below the badge.

The badge is also populated automatically each time you unlock the database, so you do not need to click **Check** after every launch.

Change Password

Click **Change Password...** to re-encrypt the SQLCipher database under a new password. The dialog asks for:

- **Current password** — must match the password the database is currently encrypted with.
- **New password** — your new passphrase. Cannot be empty and must differ from the current one.
- **Confirm new password** — must match the new password exactly.

Click **Change Password** to apply. A busy indicator and the message **“Changing password... This may take a moment.”** are shown while exif-turbo re-encrypts every page of the database. Existing **thumbnails are preserved** — exif-turbo uses a wrapped-key model, so only the wrapping key (which protects the thumbnail key) is re-encrypted under the new password; the thumbnail cache on disk does not have to be rebuilt.

Once the operation finishes successfully the dialog closes and a confirmation appears. The new password is required the next time you unlock the database.

If the current password is wrong, an inline red error message (“Current password is incorrect.”) is shown and the dialog stays open. The **Change Password...** button is disabled while indexing is in progress and while the database is locked. The operation is also rejected if thumbnail building is running when you submit the dialog.

There is no recovery if you forget the new password. Make sure you remember it (or store it in a password manager) before clicking **Change Password**.

Reset Database

At the bottom of the Settings tab, a divider separates the standard settings from a destructive-action zone.

Click the **Reset Database...** button (red) to open a confirmation dialog. Click **OK** to confirm. This permanently:

- Deletes all indexed images from the database
- Removes all indexed folder records
- Wipes the thumbnail and preview cache on disk
- Removes the per-database controlled-vocabulary term-vector index and legacy TGM compatibility snapshot

The database is vacuumed and checkpointed immediately, so the database file shrinks to near-zero on disk straight away.

While the reset runs, the modal **bulk-op progress overlay** (see [section 9](#)) covers the window and steps through each phase: *"Clearing preview cache..."* (cancelable, with a live X / Y count), *"Deleting index rows..."*, and finally *"Vacuuming database..."*. The vacuum phase cannot be interrupted — the overlay hides its **Cancel** button and shows a *"This step cannot be canceled..."* notice until it finishes.

This action cannot be undone. After a reset you will need to re-add your folders and run a full rescan to rebuild the index. Adjacent tagging sidecars are not deleted; rescanning imports them again. Rebuild Wikidata proposal vectors when needed. Existing image AI vector files are not explicitly deleted; run **AI Full Scan** when you need to rebuild them.

The **Reset Database...** button is disabled while indexing is in progress.

12. Keyboard Shortcuts

Shortcut	Action
Enter	Run search
Ctrl+F	Open / close find-in-metadata bar
Escape	Close the find-in-metadata bar
F3	Jump to next match in metadata
Shift+F3	Jump to previous match in metadata
Ctrl+T	Open / close the tagging drawer in Search or Browse

Space	Toggle the mark on the focused Search or Browse image
↓	Select the next result (Search tab) / next image (Browse tab)
↑	Select the previous result (Search tab) / previous image (Browse tab)
Page Down	Jump one page forward in results (Search tab) / Browse image list
Page Up	Jump one page backward in results (Search tab) / Browse image list
Ctrl+Q	Exit the application (File → Exit)

13. FAQ

Q: Why does the status bar say "Indexing..." even after I switch tabs?

A: The indexer runs in the background across all tabs. The pulsing blue dot in the status bar lets you know it is still working. You can continue searching and browsing while indexing proceeds.

Q: I closed the app mid-scan. Will I lose my progress?

A: No. exif-turbo saves the queue state when it closes. Next time you unlock the database, any interrupted scans are automatically resumed.

Q: The search finds nothing even though I can see files in the folder.

A: The files must be indexed first. Go to the **Indexed Folders** tab, add the folder, and click **Scan**.

Q: Does exif-turbo modify my image files?

A: Tagging, indexing, and derivative generation never modify originals. **Delete Marked Images** is an explicit destructive action that permanently deletes the selected original files after confirmation.

Q: What image formats are supported?

A: JPEG, PNG, TIFF, HEIC, BMP, GIF, and RAW formats: CR2, CR3, NEF, ARW, DNG, ORF, RW2, PEF, RAF, RWL, SRW (and any other format that ExifTool can read). Video files are also indexed: MP4, MOV, AVI, MKV, WMV, M4V, MTS, M2TS, 3GP, WebM, FLV. Thumbnails and previews for video files are extracted via PyAV (FFmpeg) — the embedded thumbnail is used when available, otherwise a frame at 1/3 of the duration. Rotation is applied so portrait phone clips render upright.

Q: Where is the database stored?

A: By default at `~/.exif-turbo/data/index/index.db` on all platforms.

Q: How do I change the database password?

A: Open the **Settings** tab and click **Change Password...** Enter your current password and a new one — exif-turbo will re-encrypt the database in place. Thumbnails are preserved (no re-indexing required). See [Change Password](#) for details.

Q: Thumbnails are not showing / are slow to appear.

A: Thumbnails are generated in a background thread after indexing. Depending on the number of images it may take a few minutes for all thumbnails to be built and cached. They persist across sessions so subsequent launches are fast.

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